
accenture
AI reinvention made real
Accenture Innovation Challenge 2026

Team details

TEAM NAME: AceX	



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Year of graduation: 2028



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Describe the problem statement (200 words)

The Problem: The Domino Effect on the Factory Floor

Think of a vehicle assembly line like a massive set of dominoes. If there's a hiccup at just one station, it creates a ripple effect down the entire line, cars pile up, waiting times spike, and mistakes happen.

The frustrating part?

Factory teams usually don't spot these issues until the damage is already done and production has slowed to a crawl. To make matters worse, plant managers are often flying blind; not every station has reliable sensors, leaving critical gaps in what they can actually see happening on the floor.

The Challenge: A Smarter Digital Twin

This is exactly what the DigitalTwin.ai challenge aims to solve. The goal is to build a live, data-driven virtual replica of the factory floor. But it's not just about making a pretty 3D dashboard.

The real challenge is creating a system that understands cause and effect. It needs to see how all the different stations interact and predict how a minor delay in one corner of the factory could snowball into a massive bottleneck or defect down the line.

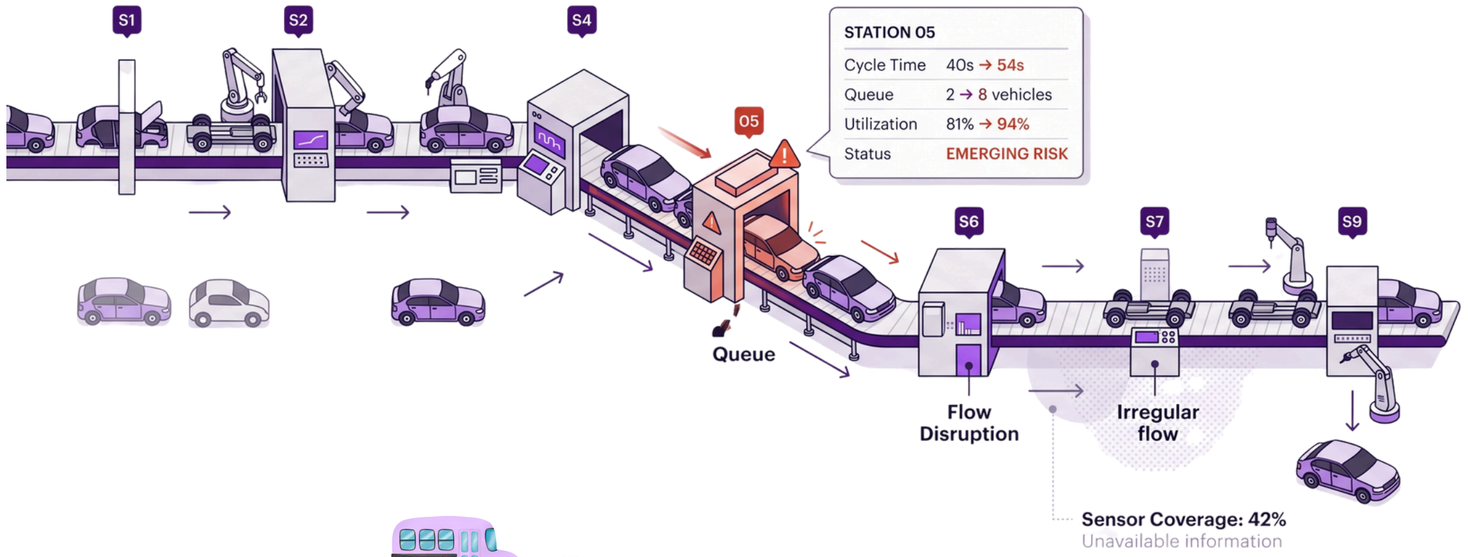
The Goal: Real-World, Actionable Intelligence

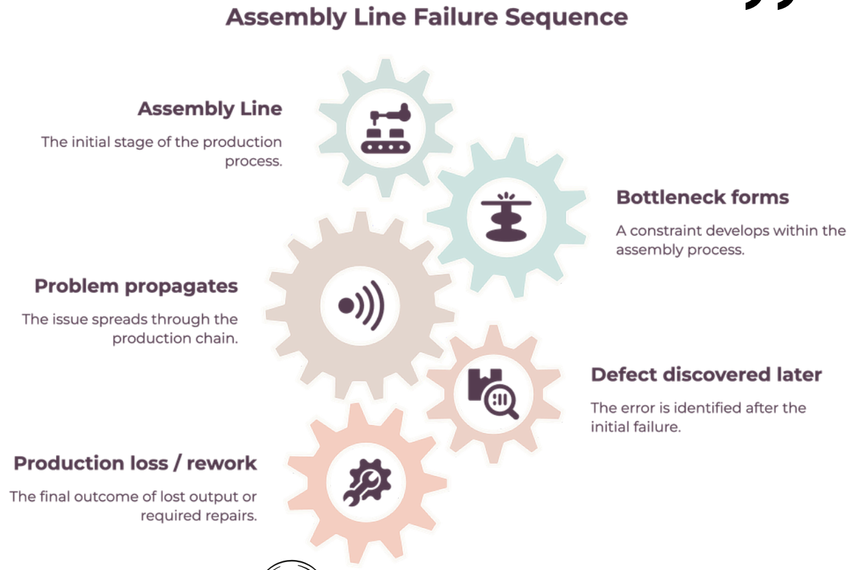
Know when to worry: It needs to tell the difference between a normal, everyday hiccup and a serious anomaly. Nobody wants an alarm that cries wolf over every little variation.

Connect the dots: When a problem does pop up, the system must investigate the root cause by looking at both the specific station and the overall flow of the factory.

Fill in the blanks: It has to work seamlessly even when sensor data is missing or incomplete.

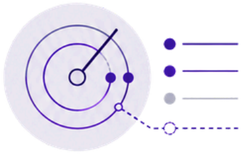
Empower the team: Ultimately, it needs to process data efficiently and give the human workers on the floor clear, actionable heads-ups before a bottleneck or defect ruins the day's production.





Proposed solution (200 words)

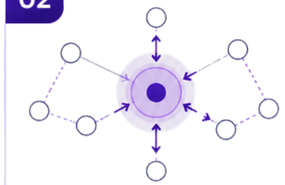
01



ADAPTIVE

Monitor what matters.
Investigate when needed.

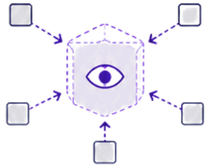
02



NETWORK-AWARE

Understand how
disruptions propagate.

03



SENSOR-RESILIENT

Infer hidden states from
surrounding signals.

A Smart, Living Digital Twin

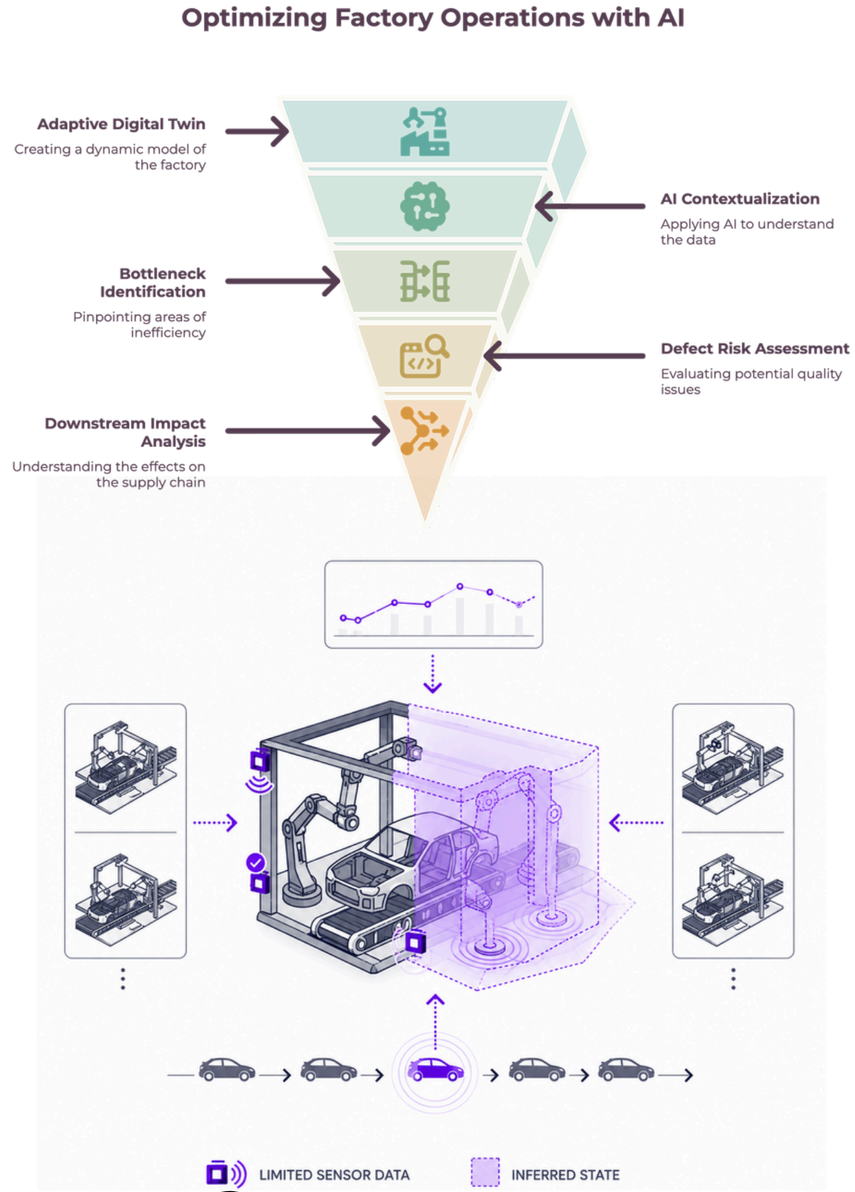
We're proposing an "Adaptive Digital Twin", a virtual, breathing replica of your assembly line that is constantly talking to the physical factory floor. By blending real-time data (like sensor readings and quality checks) with historical knowledge, it learns exactly what a healthy, normal day of production looks like.

Working Smarter, Not Harder

- Instead of drowning plant managers in endless streams of data, our system is highly strategic.
- **Focusing on vital signs:** It constantly monitors the most important health metrics of the line. It only triggers deep, heavy data analysis when it senses something might be going wrong.
- **Seeing the big picture:** It maps out the factory like a living ecosystem. It doesn't just watch isolated machines; it understands exactly how they connect and how a delay at Station A will eventually impact Station B.
- **Playing detective:** Got a blind spot on the floor with missing or unreliable sensors? No problem. The system uses clues from neighboring machines, historical trends, and the overall flow of the line to accurately guess what's happening in those dark zones.

From Putting Out Fires to Preventing Them

- By using smart algorithms, the system can easily tell the difference between a normal, everyday speed bump and a genuine bottleneck. It essentially looks into the future to predict pile-ups and defects before they have a chance to mess up production.
- **But the best part?** It doesn't just sound the alarm. It translates those predictions into clear, actionable advice for the floor team, telling them exactly where to inspect or how to shift the workload. Ultimately, it gives factory teams the power to stop reacting to problems after the fact and start preventing them before they even happen.



Video

Video Pitch Link:

https://drive.google.com/drive/folders/1LO5-deO5etwlCX3vzYrLJnHoRXOuPBtY?usp=sharing

Thank you

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